

Asymmetric Complexity and the Limits of Engine Evaluation

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1 Introduction

Chess is not a boring game. Watching two players sit silently in front of an arrangement of pieces on an 8x8 square board might seem uneventful at first. However, its enduring popularity attests to the opposite. Why is a black and white board, with some scattered pieces, so captivating?

As a player and spectator, I offer one answer: complexity.

The advent of computers (or engines) to numerically evaluate chess positions makes it easy to see who is better given a position. However, as this paper shall explore, computer evaluation on its own hides the importance of complexity in human (i.e. imperfect) chess.

1.1 The Language of Today's Chess

It is not debatable that computers are better than humans at chess. Part of this innovation is that computers can quantify the state of a position with an evaluation with centipawns as units. The sign of this evaluation dictates which side is better (+ for white, - for black), and the magnitude is analogous to being up pieces. For example, +1.0 is an advantage for the white side analogous to having one more pawn than black.

Figure 1: **Eval Bar.** This is an image from a game of mine played in the open chess platform called Lichess[1]. The engine’s evaluation is -3.3, which is reflected in the eval bar that has a larger black portion. Stockfish[2] is the chess engine Lichess uses.



If you watch any chess tournament broadcast, beside every game you will find a computer evaluation bar (see Figure 1). What the eval bar does is show which side (white or black) has an advantage if perfect play is assumed. A positive number says white is better, a negative number says the opposite.

1.2 What an Engine’s Evaluation Hides

- **Perfection is assumed:**

While a computer’s evaluation answers the question: who is better under perfect play? A spectator wants to know: who is more likely to win? Since humans are not perfect players, and far from it, a computer’s evaluation hides how hard a position is to play.

- **Difficult for whom?**

Like any competitive game, it is not always equally hard to play for both sides. For instance, one side might have an easy time attacking, while the other side has a hard time defending. In such a situation, saying that the position is difficult is not enough! One must ask: difficult for whom?

1.3 Complexity and Evaluation

- **Why is computer evaluation not enough?**

As a study tip, it is commonly suggested that players examine positions

without looking at the computer evaluation. Consider the following comparison for why this is important: suppose a position is evaluated at +3.0 by a computer, but the moves required to maintain that advantage for white are few and extremely difficult to find. Now consider another position also evaluated +3.0, where white has many options to maintain their advantage. These two positions, both of +3.0 evaluation, are not the same! The computer evaluation hides the increased possibility of making a mistake.

- **Revising Complexity:**

The fact that a position can be easy for one side and hard for another will be referred to as asymmetric complexity. Examining how asymmetric complexity revises our understanding of evaluation is the matter we shall address.

2 Procedure

Following Mark Meerschaert's [3] math modeling guidelines, we will break down the process into five steps: (1) Clarify the question (2) Select the modeling approach (3) Formulate the model (4) Solve the model (5) Answer the question and *revise if necessary*:

2.1 Clarify Question:

What does asymmetrical complexity reveal about the limits of computer evaluation?

2.2 Select the modeling approach:

Empirical, data-driven approaches are used in the academic literature on chess, trying to quantify variables of a chess position [4,5,6]. However, such an approach would be limited since there is no clear way to measure complexity. Given the dynamic relation between complexity and evaluation, our initial model will be a linear 2D dynamical system. The process for analyzing dynamical systems was taken from online lectures [7] or class notes [8].

Note: Linearity is chosen for simplicity; non-linear extensions will be discussed later.

Formulating and solving the model require sections on their own.

2.3 Formulating the linear model

- **Variables:**

E Evaluation where the sign of E indicates which side has an advantage (+ for white and - for black), and the magnitude refers to the significance of the advantage. This is the metric a computer is able to calculate.

C Complexity, how easy or hard it is for a player to play accurately. The sign indicates the net effect (+ means it is easier for white to play, - means the opposite)

$\frac{dE}{dt}$ or $\dot{E}$ Change of evaluation over time

$\frac{dC}{dt}$ or $\dot{C}$ Change of complexity over time.

Note: That units remain ambiguous should not be a problem since what matters are the dynamics between complexity and evaluation, not the empirical use of these equations.

- **General Equations:**

$$\frac{dE}{dt} = \alpha \cdot E + \beta \cdot C \quad (1)$$

$$\frac{dC}{dt} = \gamma \cdot E + \delta \cdot C \quad (2)$$

Which can be usefully written as a matrix multiplication for further investigation.

$$\begin{pmatrix} \dot{E} \\ \dot{C} \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} E \\ C \end{pmatrix} \quad (3)$$

Let the 2x2 matrix be called L.

- **Assumptions:** We must assume properties of the entries of L using chess intuitions, otherwise the model would be too general.

$\alpha > 0$: An advantage tends towards being converted.

$\beta > 0$: The rate at which an evaluation changes is positively proportional to how much the complexity favors that side. Intuition: if a position's asymmetric complexity favors white, it makes sense that the evaluation changes in favor of white.

$\gamma = 0$: The rate at which complexity changes is not related to the current advantage since for one side to be better does not necessarily mean that the position's complexity will shift in any direction.

$\delta < 0$: Complexity that favors one side tends toward 0, i.e. toward simplifying.

Note: The linearity makes sense since $(E, C) = (0, 0) \Rightarrow (\dot{E}, \dot{C}) = (0, 0)$, which would reflect a draw.

Adding the assumptions to equation (3) our model becomes:

$$\begin{pmatrix} \dot{E} \\ \dot{C} \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ 0 & \delta \end{pmatrix} \begin{pmatrix} E \\ C \end{pmatrix}, \alpha > 0, \beta > 0, \delta < 0. \quad (4)$$

2.4 Solve the model

- **Eigenvalues.** Since L is upper triangular, its characteristic polynomial factors immediately:

$$\det(L - \lambda I) = \det \begin{pmatrix} \alpha - \lambda & \beta \\ 0 & \delta - \lambda \end{pmatrix} = (\alpha - \lambda)(\delta - \lambda) = 0, \quad (5)$$

so the eigenvalues are the diagonal entries,

$$\lambda_1 = \alpha > 0, \quad \lambda_2 = \delta < 0. \quad (6)$$

By our assumptions $\alpha > 0 > \delta$, so the eigenvalues are real, distinct and of *opposite sign*. The origin is therefore a saddle point: attracting along one direction and repelling along the other.

- **Eigenvector for $\lambda_1 = \alpha$.** Solving $(L - \alpha I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 0 & \beta \\ 0 & \delta - \alpha \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \beta v_2 = 0. \quad (7)$$

Since $\beta > 0$ we require $v_2 = 0$, while v_1 is free. Taking $v_1 = 1$,

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad (8)$$

so the unstable direction is the E -axis, $C = 0$.

- **Eigenvector for $\lambda_2 = \delta$.** Solving $(L - \delta I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} \alpha - \delta & \beta \\ 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies (\alpha - \delta)v_1 + \beta v_2 = 0. \quad (9)$$

Since $\alpha > 0 > \delta$ we have $\alpha - \delta > 0$, and setting $v_1 = \beta$ gives $v_2 = \delta - \alpha$:

$$\mathbf{v}_2 = \begin{pmatrix} \beta \\ \delta - \alpha \end{pmatrix}, \quad (10)$$

which spans the line

$$C = \frac{\delta - \alpha}{\beta} E, \quad (11)$$

This is the exact line that marks the balance between advantage and complexity, that allows a game to be balanced even though advantage may be non-zero. Points on this line move toward the origin, i.e. they are draws.

- **General solution.** Because $\gamma = 0$, the second equation can be integrated:

$$\dot{C} = \delta C \implies C(t) = C_0 e^{\delta t}. \quad (12)$$

Substituting into the first equation gives the linear ODE

$$\dot{E} - \alpha E = \beta C_0 e^{\delta t}, \quad (13)$$

Note: this gives us a constant coefficient, first-order ODE of the form $\frac{dy}{dt} + p(t)y = q(t)$ which we can solve by multiplying the equation by the integrating factor $I = e^{\int p(t)dt}$.

In our case: $I = e^{\int -\alpha dt} = e^{-\alpha t}$. We can now apply the product rule.

$$\frac{d}{dt}(Ee^{-\alpha t}) = \beta C_0 e^{(\delta-\alpha)t}. \quad (14)$$

Integrating:

$$Ee^{-\alpha t} = \frac{\beta C_0}{\delta - \alpha} e^{(\delta-\alpha)t} + K \quad (15)$$

Where K is a constant. Now we can apply $E(0) = E_0$, and solve for K. Keeping in mind that α is positive, and δ negative, we have the following equations:

$$E(t) = \underbrace{\left(E_0 + \frac{\beta C_0}{\alpha - \delta}\right) e^{\alpha t}}_{\text{growing}} - \underbrace{\frac{\beta C_0}{\alpha - \delta} e^{\delta t}}_{\text{decaying}}, \quad C(t) = C_0 e^{\delta t}. \quad (16)$$

Equivalently, using our eigenvectors,

$$\begin{pmatrix} E(t) \\ C(t) \end{pmatrix} = c_1 e^{\alpha t} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 e^{\delta t} \begin{pmatrix} \beta \\ \delta - \alpha \end{pmatrix}. \quad (17)$$

- **What does equation (16) show?** The growing term vanishes precisely when

$$E_0 = -\frac{\beta C_0}{\alpha - \delta} \iff C_0 = \frac{\delta - \alpha}{\beta} E_0, \quad (18)$$

which is exactly the line spanned by $\mathbf{v}_2$. Hence, all points on this line decay to the origin, every other point eventually grows like $e^{\alpha t}$ along $\pm \mathbf{v}_1$, so that $C \rightarrow 0$ and $E \rightarrow \pm\infty$. Conveying the same result as (11), this line shows how complexity can balance evaluation such that a position where evaluation says white is better, but complexity favors black just enough, should be a draw.

- **Nullclines** Setting each derivative to zero:

$$\dot{E} = 0 \iff \alpha E + \beta C = 0 \iff C = -\frac{\alpha}{\beta} E \quad (\text{flow is vertical}), \quad (19)$$

$$\dot{C} = 0 \iff \delta C = 0 \iff C = 0 \quad (\text{flow is horizontal}). \quad (20)$$

Note that the $\dot{C}$ -nullcline $C = 0$ coincides with $\mathbf{v}_1$ which means that a trajectory starting with $C = 0$ never leaves it. This is not the case with the $\dot{E}$ -nullcline; trajectories cross it vertically.

- **Comparing slopes between the $\dot{E}$ -nullcline and $\mathbf{v}_2$:** Since $\delta < 0$ we have $\alpha - \delta > \alpha$, hence

$$\left| \frac{\delta - \alpha}{\beta} \right| > \left| -\frac{\alpha}{\beta} \right|. \quad (21)$$

Both lines have negative slope and go through the origin, but $\mathbf{v}_2$ is strictly steeper than the E -nullcline. **Note**, $\mathbf{v}_2$ is what is known as a *separatrix*, which splits regions with different behaviors. If we look at the slope of $\mathbf{v}_2$, we can identify an intuitive phenomenon: the greater β is, the less steep the separatrix becomes. We can now accurately sketch the phase portrait.

Figure 2: **Eigenvector spans and the $\dot{E} = 0$ nullcline.** The nullcline helps us visualize the trajectories.

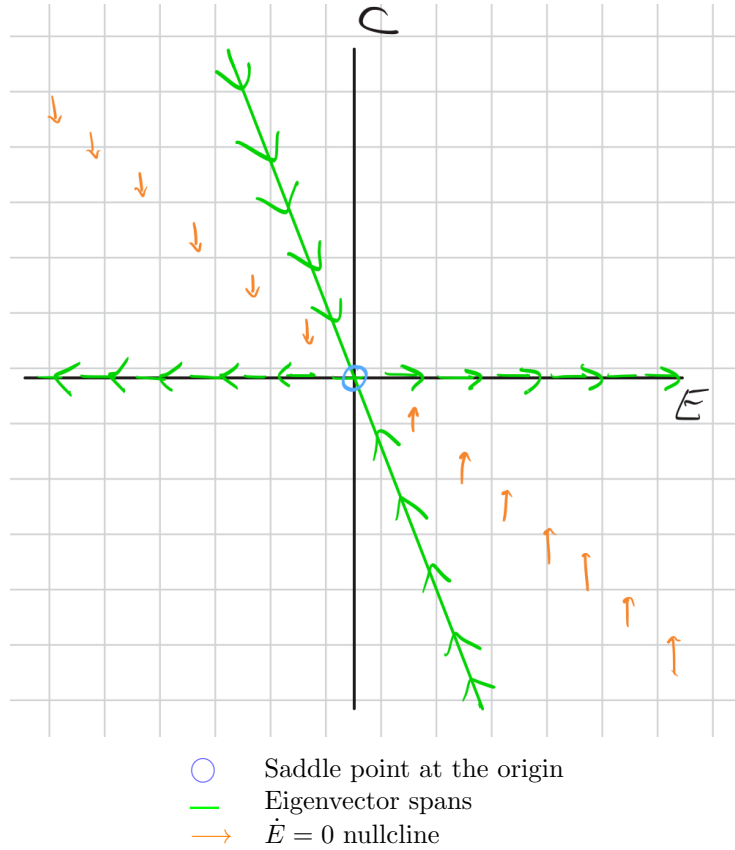
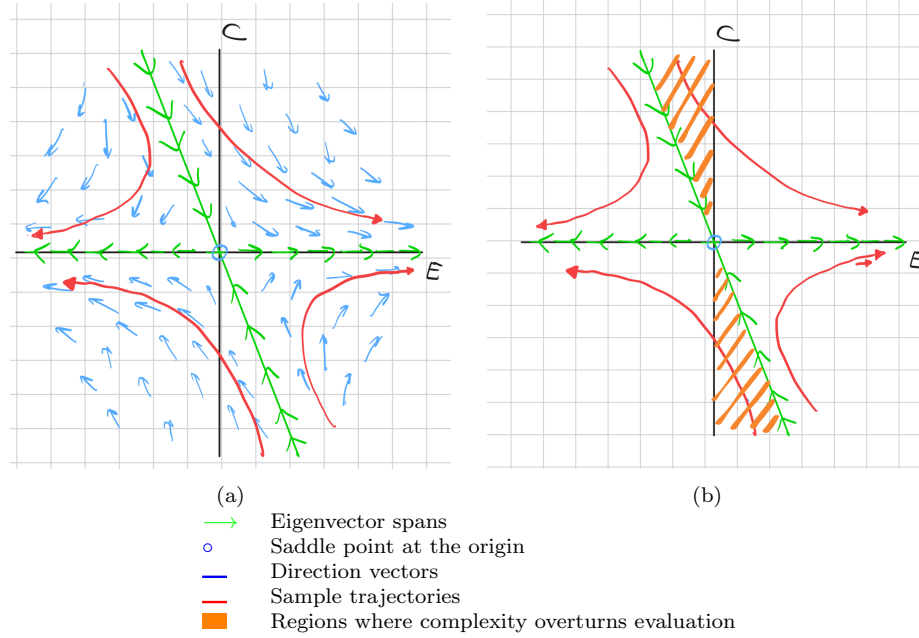


Figure 3: (a) Phase plane portrait of system (4) in the (E, C) coordinates. The origin is a saddle: trajectories are attracted along the stable eigendirection $\mathbf{v}_2$ and repelled along the unstable eigendirection $\mathbf{v}_1$. (b) Shows the interesting regions in quadrant II and IV demonstrating what evaluation, without taking complexity into account, fails to capture



Answer the question: What does asymmetrical complexity reveal about the limits of computer evaluation?

- **Key discovery:**

Computer evaluation on its own fails to capture the regions where complexity can overturn evaluation. If the complexity axis were removed, evaluation would not recognize the points in the Figure 3(b) shaded region. Let's look at the shaded regions of Figure 3 (b).

- **Shaded region in quadrant II:** For this region this model demonstrates that an asymmetrical complexity makes it so trajectories starting at a negative evaluation, but a positive complexity above the $\mathbf{v}_2$ line, end going towards a $E \rightarrow +\infty$. This is a mathematical demonstration that if black is better according to the computer evaluation, but the complexity favors white enough, then black (in human chess) is not actually better!
- **Shaded region in quadrant IV:** In this region, the same phenomena with opposite colors is demonstrated; when white has an

advantage according to a computer evaluation, but the complexity favors black enough, the result is a win for black.

- **What moves the separatrix?** Returning to the interpretation in (21), the greater β is, the less steep the separatrix is, meaning that there is a greater region where complexity may overturn evaluation. This makes sense since β refers to the influence complexity has on the change of evaluation over time. Also, if $\alpha \approx \delta$ the same phenomena occurs where the shaded region would grow. However, since $\alpha > 0, \delta < 0$ then $\alpha \approx \delta$ means that they are close to 0 while being of different signs. This means that if complexity and evaluation exert less influence on the change of complexity and evaluation respectively, the shaded region will grow. Which makes sense since it would mean that complexity would take longer to simplify, and evaluation would have less of an influence, allowing complexity to overturn the evaluation.
- **What does this mean?** This means that the effect of complexity on evaluation in human (imperfect) chess, makes computer evaluation an imperfect measurement for human advantage since it fails to consider how hard or easy it is for a human to convert an advantage.
- **Revising model:** Describing complexity and evaluation in this linear, planar model is undoubtedly simplistic. While it is an initial model which shows the basic phenomenon, further extensions should be considered. A few arise:
 1. $\gamma \neq 0$: What if γ is nonzero such that complexity varies with regard to evaluation?
 2. Non-linearity: What would a non-linear model look like? neither evaluation nor complexity extend off to infinity; an evaluation must resolve itself either at a win for black, a win for white, or a draw. Complexity must have some sort of non-linear profile since the rate at which positions become simpler is not the same when the position is slightly complex as to when it is extremely complex.

Following the same structure as before, I shall propose a non-linear model and point out important avenues for further exploration.

3 A Non-linear Extension

The linear model captures the basic phenomenon, but it has two shortcomings that can be resolved. First, evaluation should not extend to infinity: a game resolves itself as a win, a loss, or a draw, which are definitive states. Second, complexity had no profile of its own; it simply decayed at a constant rate. The extension proposed here addresses both, first formulating and then visualizing the system.

3.1 Formulate the non-linear model

Maintaining the assumptions for α, β, δ , and incorporating the non-linear extensions,

$$\dot{E} = \alpha E \left(1 - \frac{E^2}{E_{\max}^2} \right) + \beta C, \quad \alpha > 0, \beta > 0, E_{\max} = 10, \quad (22)$$

$$\dot{C} = \gamma E + \delta \sin\left(\frac{C\pi}{10}\right), \quad \gamma \geq 0, \delta < 0. \quad (23)$$

Figure 4: **Phase lines.** (left) $\dot{E}$ against E when $C = 0$: the origin is unstable and $E = \pm 10$ are stable. (right) $\dot{C}$ against C when $E = 0$: the origin is stable and $C = \pm 10$ are unstable.



Justifying new equations:

Evaluation saturates. $E(1 - E^2/E_{\max}^2)$ is similar to a logistic growth term with carrying capacity $E_{\max}$. An evaluation of ± 10 is the analog of a win or a loss: once reached, $\dot{E} = 0$ at $C = 0$ and the advantage has nowhere further to go. Refer to Figure 4 (left)

Complexity has a profile. In the linear model, $\dot{C} = \delta C$ made complexity decay at a rate proportional to itself. The sine term keeps the sign of δ (positions tend to simplify) but changes *how fast* depending on how complex the position is. At $E = 0$, $\dot{C} = \delta \sin(C\pi/10)$ vanishes at $C = 0$ and at $C = \pm 10$; simplification is fastest at $|C| = 5$ and slows to a halt as $|C| \rightarrow 10$. The chess reading is that maximally complex positions are not easily simplified, but complexity still tends towards simplifying. Figure 4 (right) shows this profile.

Complexity is bounded. Similar to the reasoning on evaluation, it did not make much sense to have infinitely complex positions. Hence, the complexity profile is such that at $C = \pm 10 \Rightarrow \dot{C} = 0$ (Refer to Figure 4 (right)).

The role of γ . In the linear model we set $\gamma = 0$. We now let it vary. When $\gamma = 0$, complexity evolves independently of evaluation and the end behavior is very close to the linear model. When $\gamma > 0$, an advantage generates complexity, the unstable curve is no longer the line $C = 0$, and, as we shall see, the number of equilibrium points changes. That is the signature of a bifurcation.

Parameter values. To draw the phase portraits we need numbers. Taking the simplest values consistent with the assumptions, $\alpha = \beta = 1$ and $\delta = -1$, with $\gamma \in \{0, 1\}$. Nothing qualitative below depends on this choice.

3.2 Solve the non-linear model

- **Nullclines.** The $\dot{E}$ -nullcline is

$$\dot{E} = 0 \iff C = -\frac{\alpha}{\beta} E \left(1 - \frac{E^2}{100} \right). \quad (24)$$

This is no longer a line but an S-shaped cubic. As before, it marks the points where complexity exactly balances evaluation so that the advantage does not move. The $\dot{C}$ -nullcline is

$$\dot{C} = 0 \iff \begin{cases} C \in \{0, \pm 10\}, & \gamma = 0, \\ E = -\frac{\delta}{\gamma} \sin\left(\frac{C\pi}{10}\right), & \gamma > 0. \end{cases} \quad (25)$$

When $\gamma = 0$ the nullcline is three horizontal lines, the middle one coinciding with the E -axis exactly as in the linear model. When $\gamma > 0$ it becomes a sine wave of amplitude $|\delta|/\gamma$ in the E direction.

- **Equilibrium points.** Equilibria are the intersections of the two nullclines. To classify them we linearize. The Jacobian of the system is

$$J(E, C) = \begin{pmatrix} \alpha \left(1 - \frac{3E^2}{100} \right) & \beta \\ \gamma & \frac{\delta\pi}{10} \cos\left(\frac{C\pi}{10}\right) \end{pmatrix}; \quad (26)$$

an equilibrium is a saddle if at that point $\det J < 0$, and a stable node if $\det J > 0$ and $\text{tr } J < 0$.

Case $\gamma = 0$. Substituting $C = 0, \pm 10$ into the $\dot{E}$ -nullcline gives $-E(1 - E^2/100) = 0, \pm 10$. The equilibria when $C = 0$ are

$$(0, 0), \quad (10, 0), \quad (-10, 0). \quad (27)$$

When $C = 10$,

$$-E \left(1 - \frac{E^2}{100} \right) = 10 \iff E^3 - 100E - 1000 = 0 \Rightarrow E \approx 13.25, \quad (28)$$

the root, solved graphically. Symmetry of this odd function means that $C = -10$ gives $E \approx -13.25$. This adds two new equilibrium points:

$$(-13.25, -10), \quad (13.25, 10). \quad (29)$$

We can now linearize near each equilibrium point to see if they are stable or unstable. At the origin,

$$J(0, 0) = \begin{pmatrix} \alpha & \beta \\ 0 & \delta\pi/10 \end{pmatrix}, \quad \lambda_1 = \alpha > 0, \quad \lambda_2 = \frac{\delta\pi}{10} < 0, \quad (30)$$

a saddle. The balanced position is still unstable. At $(\pm 10, 0)$,

$$J(\pm 10, 0) = \begin{pmatrix} -2\alpha & \beta \\ 0 & \delta\pi/10 \end{pmatrix}, \quad \lambda_1 = -2\alpha < 0, \quad \lambda_2 = \frac{\delta\pi}{10} < 0, \quad (31)$$

two stable nodes. For $(\pm 13.25, \pm 10)$, at $C = \pm 10$ we have $\cos(C\pi/10) = -1$, so with $\gamma = 0$

$$J(\pm E, \pm 10) = \begin{pmatrix} \alpha \left(1 - \frac{3E^2}{100}\right) & \beta \\ 0 & -\frac{\delta\pi}{10} \end{pmatrix}, \quad \det J = -\frac{\alpha\delta\pi}{10} \left(1 - \frac{3E^2}{100}\right) < 0, \quad (32)$$

The sign of the determinant is given by $\delta < 0$, $E^2 > 100$, $\alpha = \beta = 1$, $\delta = -1$ and $E \approx 13.2472$. Since E is always squared, $-E$ has the same determinant, so both points are saddles.

Case $\gamma = 1$. Substituting the $\dot{C}$ -nullcline $E = \sin(C\pi/10)$ into the $\dot{E}$ -nullcline gives

$$\sin\left(\frac{C\pi}{10}\right) \left(1 - \frac{1}{100} \sin^2\left(\frac{C\pi}{10}\right)\right) = -C. \quad (33)$$

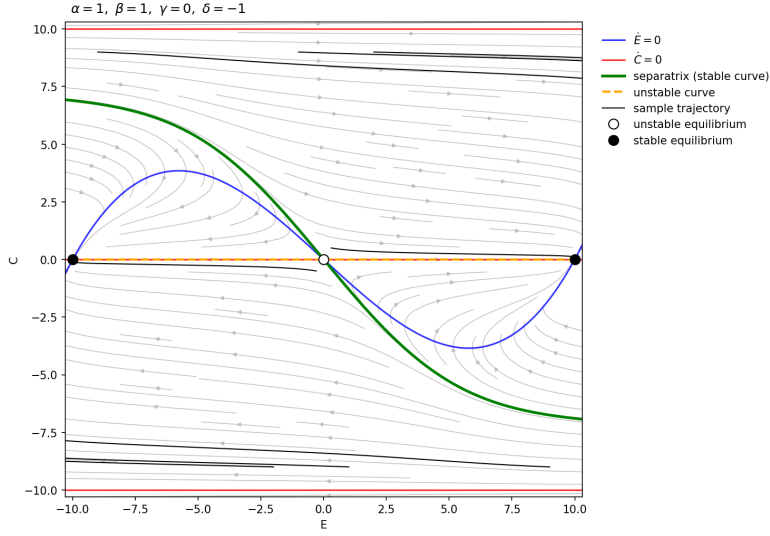
On $(-10, 10)$ the left side has the sign of C and the right side the opposite sign, so the only solution is $C = 0$. The origin is the unique equilibrium, with

$$J(0, 0) = \begin{pmatrix} 1 & 1 \\ 1 & -\pi/10 \end{pmatrix}, \quad \text{tr } J \approx 0.69, \quad \det J \approx -1.31, \quad (34)$$

again a saddle. However, the other stable points have disappeared; a bifurcation is taking place. We will prioritize the $\gamma = 0$ case since it has our desired behavior.

- Phase portraits and the separatrix. In the linear model the separatrix was a straight line, the span of $\mathbf{v}_2$. In this non-linear model it is a curve that can be calculated using a graphical solver.

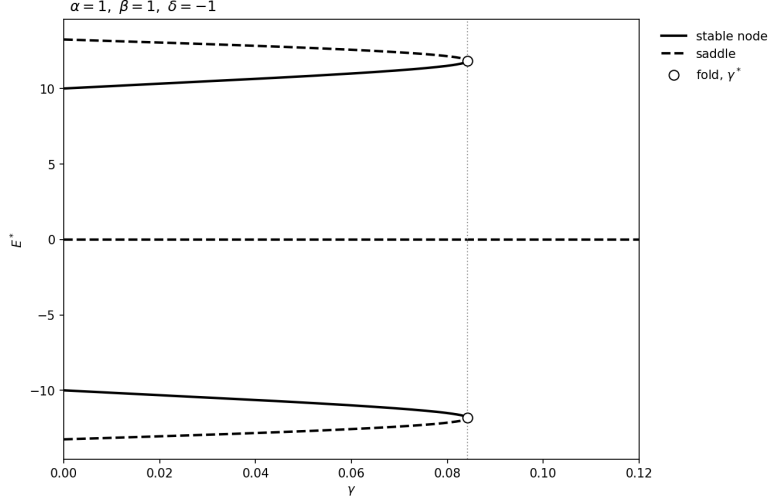
Figure 5: **Phase portrait**, $\gamma = 0$. Two stable nodes at $(\pm 10, 0)$ and a saddle at the origin. The separatrix (green) divides the plane into the regions that end in a white win and a black win. The unstable curve coincides with the E -axis, as in the linear model. Information on the tools used for plotting can be found in Appendix A.



Every trajectory off the separatrix ends at one of the two nodes (See Figure 5). The game resolves: evaluation saturates at ± 10 while complexity dies out. The origin, as well as the points on the separatrix, are the unstable points which lead to a draw (i.e. the origin).

- **A bifurcation diagram: the relation between evaluation and complexity.** Between $\gamma = 0$ (two win states) and $\gamma = 1$ (none) something must happen to the equilibria. To see what, we fix $\alpha = \beta = 1$, $\delta = -1$ and follow each equilibrium as γ increases, solving the nullcline equations numerically and plotting the resulting E^* against γ .

Figure 6: **Bifurcation diagram in γ .** Solid branches are stable nodes, dashed branches are saddles. The win-node and the outer saddle approach each other and disappear at $\gamma^* \approx 0.084$. **Note:** there are unstable points beyond $E = \pm 10$; however, they end up being irrelevant since all trajectories with $E \in (-10, 10)$ will reach a stable node before. Information on how this was plotted can be found in Appendix A.



A saddle-node bifurcation. At $\gamma^* \approx 0.084$, the stable nodes near ± 10 disappear.

What it means. γ measures how much an advantage generates complexity. Below γ^* , games still have somewhere to rest: a win is a stable state. Above γ^* there is no stable state at all. If advantage breeds complexity faster than γ^* , the stable states in our model disappear and we get $E \rightarrow \infty$. We shall remain in the $\gamma = 0$ model, where an advantage can be converted without this bifurcation.

Other parameters. Examining how any of the parameters α, β, δ or $E_{\max}$ can be bifurcation variables can be interesting, γ was chosen here because it is the parameter the linear model set to zero.

- Back to the linear model: what is preserved. Evaluate the Jacobian at the origin:

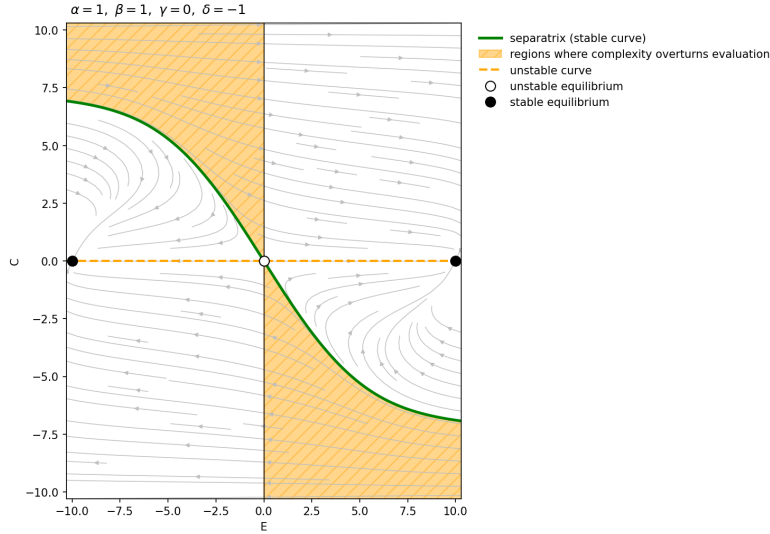
$$J(0,0) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta\pi/10 \end{pmatrix}. \quad (35)$$

With $\gamma = 0$ this is exactly the matrix L of equation (4), with δ replaced by $\delta\pi/10$. Everything derived in Section 2 about the origin carries over: it is a saddle, its unstable direction is the E -axis, and the separatrix leaves the origin tangent to

$$C = \frac{\delta\pi/10 - \alpha}{\beta} E, \quad (36)$$

the non-linear counterpart of equation (11). The linear model shows the same phenomenon as the non-linear model near the origin.

Figure 7: **Regions where complexity overturns evaluation, $\gamma = 0$.** Compare with Figure 3(b).



The shaded regions return. Figure 7 is the analog of Figure 3(b) for the non-linear model. The regions in quadrants II and IV between the separatrix and the C -axis are the same phenomenon as before: points where the computer says one side is better but the trajectory ends at the *other* side's win. Black is better according to the evaluation, the complexity favors white enough, and the game ends at $(+10, 0)$. What has changed is that the region is now bounded in C : the separatrix flattens as it approaches the invariant lines $C = \pm 10$, so there is a ceiling on how much complexity can do.

Adding to the linear answer. The non-linear model keeps the key discovery of Section 2, that computer evaluation fails to see the regions where complexity overturns it, and adds three novelties:

- 1.a *ceiling and floor* on complexity past which, no matter the evaluation, it will overturn; exactly where these limits lie is irrelevant since we are assuming the parameters;
2. *finite* win states at $E = \pm 10$, so that an advantage resolves rather than growing forever;
- 3.a *threshold* γ^* beyond which wins never settle because advantage generates complexity faster than complexity can disappear.

These differences do not affect the main phenomenon.

4 So What?

4.1 Insight into non-linear and linear dynamic systems

That the linear model effectively carried the same message as the non-linear model is not a surprising result. The separatrix and nullclines were curvy rather than straight, but the regions were analogous. This phenomenon was formalized by the Hartman-Grobman Theorem, which asserts that the linearization of a dynamical system effectively describes local qualitative patterns. While there are novelties in the non-linear model, Hartman-Grobman helps us understand why the overall dynamics are similar.

4.2 A more human evaluation

The engine's shortcoming: A computer's evaluation does not take into account *complexity*, a component that is shown to be crucial in the result of human games. The computer can calculate if a player's position is better, but if the win is only achievable if the player finds 10 obscure moves, that is a different story. Complexity is needed to differentiate between these two scenarios since asymmetric complexity matters for human chess.

Proposal for a novel metric: The problem of how exactly complexity can be calculated and how it should be used to develop a new metric is tricky. Nevertheless, from our investigation we can find some guidelines that must be followed:

1. Complexity must be asymmetric: a position can be easy for one side and hard for another; that changes the picture on who actually has better chances of winning.
2. Complexity must be combined with evaluation: complexity tells you how hard it is to play, but it does not tell you what result accurate play will lead to, that is still the role of a computer's evaluation. Finding how evaluation should be adjusted by complexity is left for further exploration.

Why this novel metric matters? Two uses this metric may have come to mind: broadcasting and education.

For broadcasting, spectators want to be able to understand what is happening in a game at a given point. If all they get is a computer evaluation, then they will not fully understand why the player with the white pieces, who is supposedly well off at +4.0, is scratching their head nervously.

For education, computer evaluation is a notoriously poor tool for learning chess. A human cares about how easy a position is to play, a computer does not. This new metric may be useful for studying strategy in chess, since making one's life easier and the opponent's life harder is a crucial part of the game.

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A Computational Tools

Digitally produced graphs were made using Python 3 [9] libraries: NumPy for linear algebra, SciPy for numerical solvers, and Matplotlib for plotting. The specific routines used were:

- `numpy.linalg.eig` — eigenvalues and eigenvectors of the Jacobian, used to classify equilibria and to obtain the stable and unstable eigendirections at the saddle.
- `scipy.integrate.solve_ivp` — Runge–Kutta (RK45) integration of the system, used for sample trajectories and for the stable and unstable curves.
- `scipy.optimize.fsolve` — locating roots as γ varied.

- `matplotlib.pyplot.streamplot` — direction field and flow lines.

Specific procedures:

- **Phase portraits (Figures 5 and 7).** The vector field $(\dot{E}, \dot{C})$ was drawn with `streamplot`, with the nullclines overlaid and sample trajectories solved using `solve_ivp`.
- **Separatrix (Figure 5 and 7).** The stable eigenvector of the Jacobian was obtained with `numpy.linalg.eig`, and was integrated with `solve_ivp`; the unstable curve was obtained identically.
- **Bifurcation diagram (Figure 6).** Starting from the known equilibria at $\gamma = 0$, `fsolve` was used at successive values of γ using the previous equilibrium as the initial guess.
- **Shaded regions (Figure 7).** The separatrix was computed as above and used as the boundary of the shaded areas, filled with `matplotlib.pyplot.fill_betweenx` between that curve and the C -axis.

B Use of AI Assistance

None of the mathematics in this project was the work of AI. Claude (Anthropic) was used only for the code described in Appendix A to facilitate plotting.